Quantum Algorithm for Linear Systems of Equations

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1. ABSTRACT & THEORETICAL FOUNDATION

We formulate the HHL algorithm for solving sparse linear systems Ax = b, achieving an exponential speedup in system dimension N, scaling as O(log(N) s2 kappa2 / epsilon).

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Solving a system with billions of equations: classical computers take days to invert the matrix, but HHL encodes the equations into quantum waves and finds the solution in seconds."

3. Mathematical Formulation & Unitary Rule

```
A|x\rangle = |b\rangle \implies |x\rangle = \frac{A^{-1}|b\rangle}{\|A^{-1}|b\rangle\|} = \fra
```

Controlled rotation by C/lambda_j effectively inverts eigenvalues in the quantum register.

4. Step-by-Step Circuit Sequence

State preparation |b>, QPE on e^{iAt}, ancilla rotation Ry(2 arcsin(C/lambda)), QPE-dagger, post-selection on ancilla |1>.

5. Executable IBM Qiskit (Python) Code

```python
from qiskit_algorithms.linear_solvers import HHL
import numpy as np

# Define 2x2 Hermitian matrix A and vector b
matrix = np.array([[1, -0.333], [-0.333, 1]])
vector = np.array([1, 1])

hhl = HHL()
print("HHL solves A|x> = |b> in logarithmic time with respect to matrix dimension.")
```

6. Computational Complexity & Speedup

Classical Time:	Quantum Time:	Speedup Class:
O(N * s * kappa) via Conjugate Gradient Method	O(log(N) * s^2 * kappa^2 / epsilon) runtime	Exponential

7. Key Applications & Future Research Directions

- Solving finite element and partial differential equations (fluid dynamics, aerodynamics)
- Quantum Support Vector Machines and big-data linear regression
- Electromagnetic scattering and structural engineering simulations

Future Work:
Quantum differential equation solvers, finite element modeling, and quantum machine learning integrations.